

CS 505 Module Seven Activity: Adjacency Matrix

Complete this template by replacing the bracketed text with the relevant information.

- Determine the adjacency matrices of r_1 and r_2 for the information provided:

Let $A_1 = \{3, 8, 4, 6\}$, $A_2 = \{6, 8, 7\}$, and $A_3 = \{1, 2, 3\}$. Let R_1 be the relation from A_1 into A_2 defined by $r_1 = \{(x, y) \mid y - x = 2\}$, and let r_2 be the relation from A_2 into A_3 defined by $r_2 = \{(x, y) \mid y - x = 1\}$.

Answer

r_1

$A_1 = \{3, 8, 4, 6\}$ & $A_2 = \{6, 8, 7\}$

Rule: $r_1 = \{(x, y) \mid y - x = 2\}$ or $y = x + 2$

$3 + 2 = 5$ is *NOT* in A_2

$8 + 2 = 10$ is *NOT* in A_2

$4 + 2 = 6$ is in A_2

$6 + 2 = 8$ is in A_2

$r_1 = \{(4, 6), (6, 8)\}$

$$r_1 = \begin{matrix} & \begin{matrix} 6 & 8 & 7 \end{matrix} \\ \begin{matrix} 3 \\ 8 \\ 4 \\ 6 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

r_2

$A_2 = \{6, 8, 7\}$ & $A_3 = \{1, 2, 3\}$

Rule: $r_2 = \{(x, y) \mid y - x = 1\}$ or $y = x + 1$

$6 + 1 = 7$ is *NOT* in A_3

$8 + 1 = 9$ is *NOT* in A_3

$7 + 1 = 8$ is *NOT* in A_3

$$r_2 = \begin{matrix} & \begin{matrix} 1 & 2 & 3 \end{matrix} \\ \begin{matrix} 6 \\ 8 \\ 7 \end{matrix} & \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{matrix}$$

2. Write pseudocode to find the shortest path between two vertices in a weighted graph using an adjacency matrix.

Dijkstra(matrix, startVertex, endVertex):

n = number of vertices

create array distance[n]

create array visited[n]

for each vertex i:

 distance[i] = infinity

 visited[i] = false

distance[startVertex] = 0

repeat n times:

 u = vertex with smallest distance
 that has not been visited

visited[u] = true

for each vertex v:

 if matrix[u][v] > 0 AND visited[v] = false:

 newDistance = distance[u] + matrix[u][v]

 if newDistance < distance[v]:

 distance[v] = newDistance

return distance[endVertex]

3. Analyze the time and space complexity of different graph algorithms when using adjacency matrix representation.

Answer:

Adjacency matrices require $O(V^2)$ space because every possible connection between vertices must be stored in the matrix, even if no edge exists.

Breadth-First Search (BFS) and Depth-First Search (DFS) both have a time complexity of $O(V^2)$ when using an adjacency matrix. This is because the algorithm must examine every possible vertex connection in each row of the matrix.

Dijkstra's shortest path algorithm also has a time complexity of $O(V^2)$ when implemented with an adjacency matrix, because it repeatedly scans the matrix to find the next shortest path.

Floyd-Warshall's algorithm, which finds shortest paths between all pairs of vertices, has a time complexity of $O(V^3)$ because it compares every vertex against every other vertex through an intermediate vertex.

The space complexity of most graph algorithms using adjacency matrices remains $O(V^2)$ due to the storage requirements of the matrix.

4. Explain whether it is possible to add the matrices provided.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

Answer:

These matrices cannot be added together. For matrices to be added, they must have the same dimensions. Matrix A is a 2×3 matrix, while matrix B is a 3×2 matrix.

5. Explain whether it is possible to multiply the matrices provided.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

Answer:

These matrices can be multiplied because their inner dimensions match. The number of columns in matrix A is equal to the number of rows in matrix B. Since A is a 2×3 matrix and B is a 3×2 matrix, the resulting matrix will be a 2×2 matrix.

6. Perform the multiplication of the matrices provided.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

Answer:

$$\begin{aligned} A \times B &= \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix} \\ &= \begin{bmatrix} (1 \times 7) + (2 \times 9) + (3 \times 11) & (1 \times 8) + (2 \times 10) + (3 \times 12) \\ (4 \times 7) + (5 \times 9) + (6 \times 11) & (4 \times 8) + (5 \times 10) + (6 \times 12) \end{bmatrix} \\ &= \begin{bmatrix} 7 + 18 + 33 & 8 + 20 + 36 \\ 28 + 45 + 66 & 32 + 50 + 72 \end{bmatrix} = \begin{bmatrix} \mathbf{58} & \mathbf{64} \\ \mathbf{139} & \mathbf{154} \end{bmatrix} \end{aligned}$$

$$A \times B = \begin{bmatrix} \mathbf{58} & \mathbf{64} \\ \mathbf{139} & \mathbf{154} \end{bmatrix}$$

7. Use matrix methods to solve the system of linear equations provided.

$$3x + 2y - z = 1$$

$$2x - y + 3z = -3$$

$$x + y + z = 2$$

$$\left\{ \begin{bmatrix} 3 & 2 & -1 & 1 \\ 2 & -1 & 3 & -3 \\ 1 & 1 & 1 & 2 \end{bmatrix} \right\}$$

RREF:

$$\left\{ \begin{bmatrix} 1 & 0 & 0 & -\frac{15}{13} \\ 0 & 1 & 0 & \frac{33}{13} \\ 0 & 0 & 1 & \frac{8}{13} \end{bmatrix} \right\}$$

$$x = -\frac{15}{13} = \sim -1.154$$

$$y = \frac{33}{13} = \sim 2.538$$

$$z = \frac{8}{13} = \sim 0.615$$

8. Explain how matrices can be used to represent graphs.

Answer:

Graphs can be represented using an adjacency matrix. In an adjacency matrix, the rows and columns represent the vertices of the graph. The value in each cell indicates whether an edge exists between two vertices. For an unweighted graph, a value of 1 typically indicates a connection and a value of 0 indicates no connection. For a weighted graph, the cell contains the edge weight instead of 1. Adjacency matrices provide a structured way to store graph data and make it easy to determine whether two vertices are connected.

9. Explain how matrix operations can be used to solve problems in computer science.

Answer:

Matrix operations can be used to solve many problems in computer science. In graph theory, matrices can represent graphs and be used to analyze connections and paths between vertices. In computer graphics, matrices are used to perform transformations such as scaling, rotating, and translating objects on a screen. Matrices are also used in machine learning and artificial intelligence to store and process large amounts of data efficiently. Additionally, matrices can be used to solve systems of linear equations, which are common in scientific computing, simulations, and optimization problems. Their ability to organize and manipulate data makes matrices an important tool in many areas of computer science.

10. Determine the determinant of A for the matrix provided.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 4 & 6 \\ 6 & 8 & 3 \end{bmatrix}$$

$$\begin{aligned} \det(A) &= 2\det\begin{bmatrix} 4 & 6 \\ 8 & 3 \end{bmatrix} - 3\det\begin{bmatrix} 1 & 6 \\ 6 & 3 \end{bmatrix} + 1\det\begin{bmatrix} 1 & 4 \\ 6 & 8 \end{bmatrix} \\ &= 2[(4 \times 3) - (6 \times 8)] - 3[(1 \times 3) - (6 \times 6)] + 1[(1 \times 8) - (4 \times 6)] \\ &= 2(12 - 48) - 3(3 - 36) + 1(8 - 24) \\ &= 2(-36) - 3(-33) + 1(-16) \\ &= -72 + 99 - 16 \\ \det(A) &= 11 \end{aligned}$$

11. Determine the inverse of A for the matrix provided.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & 4 & 6 \\ 6 & 8 & 3 \end{bmatrix}$$

$$\text{Gauss-Jordan} = \begin{bmatrix} 2 & 3 & 1 & 1 & 0 & 0 \\ 1 & 4 & 6 & 0 & 1 & 0 \\ 6 & 8 & 3 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{RREF} = \begin{bmatrix} 1 & 0 & 0 & -\frac{36}{11} & -\frac{1}{11} & \frac{14}{11} \\ 0 & 1 & 0 & 3 & 0 & -1 \\ 0 & 0 & 1 & -\frac{16}{11} & \frac{2}{11} & \frac{5}{11} \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -\frac{36}{11} & -\frac{1}{11} & \frac{14}{11} \\ 3 & 0 & -1 \\ -\frac{16}{11} & \frac{2}{11} & \frac{5}{11} \end{bmatrix}$$

12. Determine the eigenvalues and eigenvectors of the matrix provided.

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Answer:

Eigenvalues:

$$\begin{aligned} 0 &= \det(A - \lambda I) \\ &= \det\left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) \\ &= \det\left(\begin{bmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{bmatrix}\right) \\ &= (3-\lambda)(3-\lambda) - 1 \\ &= 9 - 6\lambda + \lambda^2 - 1 \\ &= \lambda^2 - 6\lambda + 8 \\ &= (\lambda - 4)(\lambda - 2) \end{aligned}$$

$\lambda = 4$ & 2

Eigenvectors:

$$\begin{aligned} \lambda &= 4 \\ 0 &= (A - \lambda I)v \\ &= \left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}\right)v \\ &= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}v \\ &= \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= -x + y \\ x &= y \end{aligned}$$

When $x = 1, y = 1$

$$\text{Eigenvector} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} \lambda &= 2 \\ 0 &= (A - \lambda I)v \\ &= \left(\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\right)v \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}v \\ &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \\ &= x + y \\ x &= -y \\ \text{When } x = 1, y &= -1 \\ \text{Eigenvector} &= \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

